

Smart Alfalfa Irrigation System for Semi-Arid Regions

Youmin Li

Department of Food & Resource Economics/IFAS

Problem Statement:

In Western semi-arid states (CA, AZ, NV, UT, CO), water conservation is critical, alfalfa is primary field crop because it's biological characters. So my idea is try to build a "smart irrigation controller" for alfalfa that measures soil moisture and microclimate (air temperature / humidity) and then automatically turns a water pump + pressure irrigation line on/off based on a moisture threshold with hysteresis, plus simple "Western semi-arid" logic (hot/dry conditions → irrigate earlier; cool/humid → irrigate less).

This system monitors real-time soil moisture and environmental conditions to activate a pressure irrigation pump only when necessary. By maintaining soil moisture within the optimal 50% depletion range, the system ensures high forage yield while minimizing water waste. This system is a small-scale smart irrigation platform designed for alfalfa grown in western semi-arid environments. It will operate in the field and use an Arduino Mega 2560 as the main microcontroller, with an ESP32 module for cloud data communication. The system will monitor soil moisture and environmental conditions, display system status on an LCD1602, and control irrigation through a 5V relay connected to a water pump. Its key functions are sensing, decision-making, pump control, display of operating mode, and remote data transmission.

I/O Architecture Description:

The soil moisture sensor connects to an analog pin on the Arduino Mega, so the controller can read its voltage signal through the ADC. The DHT11 connects to a digital input pin because it sends data digitally. The 5V relay connects to a digital output pin, and the relay then switches external power to the water pump. The LCD1602 connects through digital communication pins and displays live values and system mode. A auto/manual mode switch can connect to another digital input pin, allowing the operator to change modes. The ESP32 create own WiFi through soft-app, which can read connected with phone, receive live data and send commands to irrigation system in remote control. The ESP32 communicates with the Mega through serial pins, it uploads sensor and pump data, and receive Openweather forest data through API to the cloud platform (Thingspeak). In this architecture, the Mega receives sensor inputs, processes the control logic, drives the outputs, updates the display, and exchanges data with the ESP32.

If the moisture level drops below a preset threshold, the system energizes the relay and turns the water pump on. If the moisture level rises above the target range, the pump is turned off. The DHT11 readings provide supporting information: during hot and dry conditions, the controller may interpret low moisture as more urgent and irrigate earlier. The LCD1602 confirms the operating mode by displaying messages and gives the user immediate confirmation of system operation and field conditions.



Objective: Keep input water level maintain production and export



Description of sensors used and outputs:

Input Sensors:

Component	Sensor	Output format	Signal type	System role
Soil Moisture Sensor V2.0	Measures soil wetness in the alfalfa root zone	Analog voltage read by Arduino ADC, converted to soil moisture percentage	Active analog sensor	Primary irrigation-decision sensor; It measures soil wetness in the root zone. Connected to A0, reads analog and convert to % moisture using calibration points (dry/wet).
Water Level Sensor	Measures water level in the reservoir/tank	Analog voltage read by ADC and converted to water-level percentage	Active analog sensor	Monitoring irrigation volume for recoding each irrigation time, set up water irrigation budget according to conditions, and add water in time
DHT11 Temperature/Humidity Sensor	Measures air temperature and relative humidity	Digital data: temperature in °C and RH in %	Active digital sensor	It reads air temperature and humidity to detect high-evapotranspiration conditions (hot and dry days). Reads digital data → temperature (°C) and RH (%).
Mode Button	User mode selection, not an environmental sensor	Digital HIGH/LOW signal	Passive input	Toggles between AUTO and MANUAL mode
4×4 Keypad	User command input	Digital row/column key scan	Passive input	Sets irrigation volume and starts/stops manual irrigation

Outputs:

OUTPUT	FORMAT	FUNCTION
5V relay + pump	Digital ON/OFF output	Switches pump power. When soil moisture is below the low threshold, the relay turns the pump ON; when moisture rises above the high threshold, the relay turns the pump OFF.
LCD1602	16×2 text display	Shows AUTO/MANUAL mode, soil moisture, pump status, temperature, humidity, and irrigation volume.
ESP32 own IP dashboard	WiFi webpage dashboard	Displays live sensor and pump data on a phone through the ESP32 access point.
ESP32 data communication	Cloud data fields	Uploads DHT11 temperature, soil moisture, water level, OpenWeather temperature, irrigation volume, pump status, DHT11 humidity, and OpenWeather main forecast.

Program Flowchart/state diagram:

The system can be described as a multi-state controller with five major states: Start/Initialize, Auto Monitoring, Auto Irrigation, Manual Irrigation, and Safety/Fault Stop.

- **Automated Threshold Logic:** The system triggers irrigation when soil moisture drops below a specific threshold (simulating the 30% allowable depletion level for alfalfa).
- **Manual Irrigation Setting:** Force stop or start irrigation, set up irrigation volume and budget through pin pad.
- **Actuation:** A 5V relay module switches a DC water pump on or off based on the sensor data.
- **Safety Lock:** Arduino output pin toggles relay → pump ON/OFF. Turn ON when moisture < LOW_THRESHOLD; Turn OFF when moisture > HIGH_THRESHOLD. The system prevents over-irrigation by checking moisture levels every few seconds and turning off the pump immediately once the target saturation is reached.
- **Safety limits:** Max runtime per irrigation event (e.g., 10 minutes); Minimum rest time between events (prevents short cycling); Sensor fault detection (disconnected / out-of-range readings); Manual override button (toggle pump for testing)



Operational description:

When the system starts, the LCD will briefly show a startup message such as “Smart Irrig.” and “Mega2560 Ready” if the system working. After initialization, the user will see live field information on the LCD, including mode (auto/manual), soil moisture percentage, pump status (on/off), temperature (°C), humidity (%), and target irrigation volume (ml/L).

If the soil moisture percentage falls, temperature, humidity below the low threshold and OpenWeather data forecast no precipitation in next 3 days, the relay turns ON and activates the water pump. The LCD and dashboard should then show the pump as ON. When irrigation raises the soil moisture above the high threshold, the pump turns OFF automatically. This hysteresis prevents rapid ON/OFF switching and helps avoid relay chattering.

If temperature is high and relative humidity is low, the system increases irrigation sensitivity by adjusting the moisture thresholds upward. In practice, this means the pump may start earlier because alfalfa water demand is expected to be higher under high-evapotranspiration conditions. Your code uses hot/dry threshold logic with temperature, humidity, and a bias adjustment.

In manual mode, the user can control irrigation through the keypad or remote ESP32 commands. The user enters a target volume in milliliters, presses # to save it, and presses A to start manual irrigation. The system estimates pump runtime from the target volume and pump flow rate. Pressing B stops the pump, C forces manual mode, and D returns the system to automatic mode. These keypad functions are included in the project documentation and code logic.

The ESP32 adds a remote-monitoring layer. It receives telemetry from the Mega, hosts a local phone dashboard (SmartIrrigation-Youmin, IP: 192.168.4.1), sends commands back to the Mega, collects OpenWeather data, and uploads selected variables to ThingSpeak. The GitHub documentation states that the ESP32 runs in AP + STA mode: AP mode creates its own WiFi network for phone monitoring, while STA mode connects to home WiFi for OpenWeather and upload ThingSpeak (via write API) with live local sensor data.

Summary:

The Smart Alfalfa Irrigation System is a low-cost IoT irrigation controller designed for semi-arid alfalfa production (Mega board + ESP32 + Sensor around). It combines analog soil moisture sensing, analog water-level monitoring, digital temperature-humidity sensing, automatic pump control, local LCD feedback, keypad/manual operation, and ESP32-based dashboard/cloud communication. The Arduino Mega 2560 performs the core sensing and control functions, while the ESP32 extends the system to phone-based monitoring, OpenWeather data integration, and ThingSpeak upload.

The main contribution of this smart irrigation system: 1. Flexible and precise irrigation volume according to real-time soil moisture, crop biocharacters, water availability, and microclimate conditions; 2. Less labor input and in time irrigation once evaporation higher than precipitation; 3. Water saving and higher efficiency because accurate irrigation to water absorb area of the crop; 4. Use the historical irrigation recording data, which uploaded in cloud platform (ThingSpeak) and analyzed by supervised machine learning in Matlab, to train the irrigation system to set better and better irrigation decision. This makes the design more appropriate for western semi-arid agriculture, where alfalfa production depends heavily on efficient water use, avoiding both water stress and water waste due to non-precise irrigation practices (e.g. gravity irrigation, fixed irrigation schedule, groundwater depletion).

Reference:

Allen, R. G., Pereira, L. S., Raes, D., & Smith, M. (1998). *Crop evapotranspiration: Guidelines for computing crop water requirements* (FAO Irrigation and Drainage Paper 56). Food and Agriculture Organization.

Griggs, T. C. (Utah State University Extension). *Soil Moisture Monitoring / Irrigation scheduling for alfalfa and irrigated pasture*.

USDA Agricultural Research Service. *Develop new tools and strategies for management of alfalfa production in the Western U.S. that conserves scarce water resources*.

Utah State University Extension. *4R's of Irrigation Management*.

Appendix

Coding:

Arduino 2560 (C++): https://github.com/liyoumin/Geospatial-AgEcon/blob/main/smart-irrigation/smart_irrigation_youmin.ino

ESP32 for data communication: https://github.com/liyoumin/Geospatial-AgEcon/edit/main/smart-irrigation/ESP_data_communication.ino

Readme: <https://github.com/liyoumin/Geospatial-AgEcon/blob/main/smart-irrigation/README.markdown>

Project Video Link:

<https://liyoumin.github.io/personalweb/project/scikit/>



Coding share links:

<https://github.com/liyoumin/Geospatial-AgEcon/tree/main/smart-irrigation>